CS 830 Project Proposal
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CS 830 Introduction to Artificial Intelligence, Prof. Wheeler Ruml

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Problem statement **1**
ClaudeAI and LogicLLAMA **3**
First Order Logic as a language to represent Natural Language: **3**
Time and Space complexity, State Space **3**
How to generate a really long knowledge base and convert to Skolemized clausal forms? **3**
Example Knowledge Bases, Clausal Forms and proofs **4**
Example Knowledge base 1: 4
Skolemized Clausal forms (manual): 5
Knowledge base 2 (with noise): 6
Skolemized Clausal forms (manual): 7
What changed from previous version: **8**
Scanned copy **9**
References **10**

Problem statement

Input (unlike assignment 6): Only one knowledge base in natural language that can have multiple First Order Logic interpretations. (An interpretation is an assignment of meaning to the symbols of a formal language.)
Example1: Bat is a mammal. Bat is used in baseball.
Example2: Bark is a boat. Bark is a sheet of chocolate.
Domain of discourse: All words that have two different meanings {bat1 , bat2, bark1, bark2} where the meaning of each word is different. bat1 != bat2, bat1 != bark1, bat1!=bark2 and so on.

Output (**same as output format from assignment 7**): If there is a proof, proof's resolution steps one step per line and total number of resolutions will be printed. If there is no proof, prints No proof exists.
Brief Method summary: Input will be processed and converted to Skolemized CNF, unify over CNF and apply resolution refutation. (Unify will be part of resolution refutation).

Goal: Prover finds a contradiction between the theory and negated query by deriving the empty clause, which proves the query.
Objective: Path cost function i.e. the number of steps and correctness of the proof for two representations

Detailed Method Summary:

1) Input: knowledge base in natural language (as it was asked in the assignment 6 and 7)
2) Create a natural language knowledge base that can have multiple FOL representations, that is it can have multiple interpretations i.e. atleast 2.
3) Convert knowledge base to First Order Logic (FOL) representation 1 and 2 respectively (also without equality) (**using an existing work that I implemented a while ago**),
4) System used for generating multiple representations in FOL is a system that uses https://github.com/gblackout/LogicLLaMA and will be manually verified and edited for correctness of automatic conversion results.
5) Convert FOL to CNF (Skolemized CNF)
6) CNF to Clausal forms (Skolemized CNF)
7) Unify over Skolemized CNF
8) Apply Resolution refutation
9) Extract proof steps from the path
10) Plot graph for small Knowledge bases the time to find proof, total number of resolutions.
11) Give a prompt to LogicLLAMA and ClaudeAI using the apis as follows:
    can you prove using resolution refutation on the following set of Skolemized CNF and output as follows?
    <Resolution steps>
    .....
    <total number of resolutions>
    .....
    <time taken to find proof for this CNF input>
    .....
    <state space for this for this CNF input>
    .....
    <print full proof path>
    .....
    <print your proof plan>
    .....
    If you did not find any proof, then ignore the above format and print “No Proof exists”.

12) Plot the graph for the same for LogicLLAMA and Claude AI
13) Compare time taken, state space and proof paths.

ClaudeAI and LogicLLAMA

ClaudeAI [6] and LogicLLAMA[5] are two large language models. ClaudeAI is built on ConstitutionalAI with self-reflection and feedback loop to learn from its responses to avoid generating harmful texts. ClaudeAI follows Reinforcement Learning with AI feedback (RLAIF) [6] while LogicLLAMA [4] follows Reinforcement Learning with Human Feedback (RLHF) [5]. LogicLLAMA is an extended version of alpaca-lora and uses a specific type of input techniques. Prompts are inputs to Large Language Models (LLMs) and they can be adjusted to get required output in specific formats. The LLMs are sensitive to prompts and tend to learn in-context [5]. Therefore applying prompt specific techniques to compare to that of traditional resolution refutation results would be easier.

First Order Logic as a language to represent Natural Language:

Natural Languages are known to have inherent expressive power [1]. First Order Logic provides a mechanism for systems to better understand unstructured text towards language understanding tasks. [3] Capturing the expressive power of natural language by using First Order logic representations is another commonly followed direction [1] [2].

Time and Space complexity, State Space

W.r.t to time, only one pair at a time is resolved, when there are multiple complementary literals and two statements can get illegal resolution depending on the sequence of resolution. The order matters. When we do hand-solve, we have some clues to make it more general, but it is difficult to stop the prover as the nested functions for a single predicate can grow. The worst case size can grow exponentially and the prover can run forever. Size of history increases as the resolution explores all possible combinations. Space depends on the number of clauses and complexity of the clauses (depth of nested clause). We did not consider equality and it seems that it reduces the number of possibilities or paramodulation possibilities for this specific use case of considering equality.

How to generate a really long knowledge base and convert to Skolemized clausal forms?

1) Creating larger knowledge base and converting to clausal forms - Use LogicLLAMA and Claude AI to create a really large Knowledge base in clausal forms. And evaluating if the resulting clausal forms are correct may be challenging due to hallucinations and other known issues.[7]

Example Knowledge Bases, Clausal Forms and proofs

VS

Example Knowledge base 1:

Bat is a mammal.
Bat is used in baseball.
Bats have a barrel.
Bat flies with wings.
Wings are not barrels.
Barrels are not wings.
Bats can fly.
There is a brown bat.
--- query ---
Brown Bat can fly.
--- negated query ---
Brown Bat cannot fly.

Will derive bottom since the negated query is False and the query is True.
There is at least one bat that is brown and bats can fly.

CNF grammar, same as given in :

clause → literal | clause
| literal
literal → predicate
| - predicate
predicate → capitalized-name (term-list)
term-list → term , term-list
| term
term → capitalized-constant-name
| capitalized-function-name (term-list)
| lowercase-variable-name

Skolemized Clausal forms (manual):

1. Mammal(Bat) | Brown(Bat)
2. UsedIn(Bat, Baseball)
3. (∀x (Bat(x) -> Has(x, Barrel))
4. (∀x (Bat(x) -> ∃y (Wing(y) ∧ FliesWith(x,y))))
5. (∀x (Wing(x) -> ¬ Barrel(x))
6. (∀x (Barrel(x) -> ¬ Wing(x))
7. (∀x (Bat(x) -> CanFly(x))
8. (∃x (Bat(x) ∧ Brown(x)))

   --- query ---

   (∃x (Bat(x) ∧ Brown(x) -> CanFly(x)))

   --- negated query ---

   ¬ (∃x (Bat(x) ∧ Brown(x) -> CanFly(x)))

query:

(∃x (Bat(x) ∧ Brown(x) -> CanFly(x))) =>	∃x (¬ (Bat(x) ∧ Brown(x)) ∨ CanFly(x)))
	∃x (¬ Bat(x) ∨ ¬ Brown(x)) ∨ CanFly(x)))
Skolemization constant c	¬ Bat(c) ∨ ¬ Brown(c) ∨ CanFly(c)
Final Clausal form	{ ¬ Bat(c) , ¬ Brown(c)), CanFly(c))) }

Negated query:

(∃x (Bat(x) ∧ Brown(x) -> CanFly(x))) =>	¬ ∃x (¬ (Bat(x) ∧ Brown(x)) ∨ CanFly(x)))
Apply Quantifier Negation	∀x ¬ (¬ (Bat(x) ∧ Brown(x)) ∨ CanFly(x)))
	∀x (¬¬ (Bat(x) ∧ Brown(x)) ∧ ¬CanFly(x)))
	∀x ((Bat(x) ∧ Brown(x)) ∧ ¬CanFly(x)))
	((Bat(x) ∧ Brown(x)) ∧ ¬CanFly(x)))
Final Clausal form	{ {Bat(x) }, { Brown(x) }, {¬CanFly(x) } }

Proof:

1. Mammal(Bat) | Brown(Bat)
2. UsedIn(Bat, Baseball)
3. ¬Bat(x) ∨ Has(x, Barrel)
4. ¬Bat(x) ∨ Wing(f(x))
5. ¬Bat(x) ∨ FliesWith(x,f(x))
6. ¬Wing(x) ∨ ¬Barrel(x)
7. ¬Barrel(x) ∨ ¬Wing(x)

8. ¬Bat(x) ∨ CanFly(x)
9. ¬Bat(x) ∨ Brown(x)
10. Bat(c)
11. Brown(c)
12. ¬CanFly(c)

Resolve 8 & 10 => ¬Bat(x) ∨ CanFly(x) and Substitute x = c

¬Bat(c) ∨ CanFly(c) Bat(c)
—----------
CanFly(c)

13. CanFly(c)

Resolve 12 & 13 => CanFly(c) ¬CanFly(c)
—----------
[ ]

Therefore it entails that Brown Bat can fly.
Proof steps 4.

Example Output for this problem is:

1: ¬Bat(c) ∨ CanFly(c)
2: Bat(c)
3: CanFly(c)
4: ¬CanFly(c)
1 & 2 give 3: CanFly(c)
3 & 4 give 5: <empty>

Knowledge base 2 (with noise):

Bat is a mammal.
Bat is used in baseball.
Bats have a barrel.
Bats have wings.
Bats can fly.
Flying requires wings.
Wings are not barrels.
Barrels are not wings.
Bat can be elegant.
There is a brown bat.

--- query ---
Elegant Brown Bat has a barrel and can fly.
--- negated query ---
Elegant Brown Bat does not have a barrel or cannot fly.

Will not derive bottom since the negated query is true and the query is false.
There is atleast one bat that is brown, bats can fly, flying requires wings and wings are not barrels and barrels are not wings.

Skolemized Clausal forms (manual):

1. Mammal(Bat)
2. UsedIn(Bat, Baseball)
3. ¬Bat(x) ∨ Barrel(b(x))
4. ¬Bat(x) ∨ Has(x,b(x))
5. ¬Bat(x) ∨ Wing(w(x))
6. ¬Bat(x) ∨ Has(x,w(x))
7. ¬Bat(x) ∨ CanFly(x)
8. ¬CanFly(x) ∨ Wing(wf(x))
9. ¬CanFly(x) ∨ Has(x,wf(x))
10. ¬Wing(x) ∨ ¬Barrel(x)
11. ¬Barrel(x) ∨ ¬Wing(x)
12. Bat(e)
13. Elegant(e)
14. Bat(c)
15. Brown(c)
16. Bat(d)
17. Brown(d)
18. ¬CanFly(d)

Resolve 7 and 16
¬Bat(x) ∨ CanFly(x) and Bat(d)
Substitute x = d

¬Bat(d) ∨ CanFly(d) Bat(d)
—----------
CanFly(d)

19. CanFly(d)

Resolve 19 and 18,

```
  ¬CanFly(d)      CanFly(d)
—------------------------------------
                [ ]
```

Example Output for this problem is:

1: ¬Bat(x) ∨ CanFly(x)
2: Bat(d)
3: ¬CanFly(c)
4: CanFly(c)
1 & 2 give 3: ¬CanFly(c)
3 & 4 give 5: <empty>

What changed from previous version:

Removed a lot of stuff. It simplified and removed abstract stuff following guidelines of the professor. I may or may not complete all Large Language Models relevant work depending on time and especially computational resources as well as the fact that I was able to help getting OpenAI EDU licenses for the classes during last semester and I do not have a license for OpenAI. However I planned my work around other LLMs and I am fine.

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CS 830 Introduction to Artificial Intelligence

Prof. Wheeler Ruml

Project proposal, March 11th 2025, Student: Sushma Anand Akoju

Project design idea

I work at the intersection of natural language understanding and neurosymbolic methods for ethical, responsible and fair use of AI. I was hoping to design a knowledge base similar to the ones designed for assignments 6 and 7. I have been studying about measuring logical entailment over Language models. Since I have been working on first order logic and proofs for sometime now and more recently CS 830 has been working on resolution refutation proofs, **following are the concrete steps I had on mind:**

1) To design knowledge base in natural language (as it was asked in the assignment 6 and 7),
2) To convert knowledge base to First Order Logic (FOL) representation (using an existing work that I implemented a while ago),
3) To convert FOL to clausal form, and
4) Run unify and
5) Run resolution refutation
6) Extract proof steps from the path
7) Use the 2), 3) and 6) to run it through **LogicLLaMa, GPT, Claude AI**
8) Direct Prompt structures to use: **Chain-of-thought & Graph-chain-of-thought**
9) Compare the results about why, how and when do we say the algorithms and language model did well

what does this mean?

what will you measure?

not clear to me

The reason steps 1 through 7 are simple is because we already did the majority of the work for assignments 6 and 7, step 6 is not difficult to capture successful steps and failure cases. Step 7 is a simple prompt design which is already available. It should not take more than 5-30 minutes to record the results from language models.

The most challenging part would be step 8) - the part about reading and writing

a) the why, how and when do we conclude the algorithm did well (by appealing to time complexity, state space, search space and backtracking) versus .
b) Why, how and when do we conclude the language model did well and also attempt to score the performance.

What parts of step 8 are challenging?

1) Not challenging part - Step a) is something we do during the homework assignments and lectures discuss elaborately on this.
2) **Challenging part:** Step b) can be inferred from a few recent research works and I could summarize that to re-run to extract some additional details.

3) **Challenging part:** How to select and organize results from steps a) and b) for comparison.
4) Compare where and how step a) and step b) results would differ and why due to design, complexity, results would be the part requiring more focussed and cautious approach.
5) Cite the references of research works relevant for this.

needed for proposal

What if this project is too small and I would end up completing it earlier?
It is not a small project. But if that happens, I would consider running a moral reasoning example using Cp-nets shared by the professor. By using CP-nets approach of preference (example: https://arxiv.org/pdf/1107.0023 page 141) based conditional modeling of the knowledge graph, I would like to design the trolley problem knowledge base, following all of the aforementioned 8 steps. I am not familiar about which algorithms can be used for equivalent proofs and satisfiability, but I would consider designing the trolley problem knowledge base anyway to run through the same algorithms and **discuss why it would not work. That would allow me to learn about an unknown concept and why aforementioned algorithms would not work for trolley problems designed using CP-nets. This way I would not be overcommitting to learning CP-nets and implementing it but I would be learning why aforementioned approaches may not work for CP-nets type of data format.**

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